

ASSIGNMENT: Blockchain Tech [Afternoon]

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Chapter 11: “Business and Legal Prerequisites for Blockchain and Smart Contracts”

1. Why Consensus Exists (Centralized vs. Decentralized)

Consensus exists to make sure all participants in a system agree on the same data and state of transactions.

1) Centralized Systems

- One central authority (e.g., a bank or server) controls data.
- Trust is placed in a single organization.
- No consensus mechanism is needed because the authority decides what is correct.
- Example: A traditional bank database.

2) Decentralized Systems (Blockchain)

- No single authority controls the system.
 - Many independent nodes maintain copies of the ledger.
 - Consensus is required so all nodes agree on:
 - Which transactions are valid
 - The order of transactions
 - This prevents fraud, double spending, and manipulation
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2. Safety, Liveness, and Finality

These are core properties of blockchain consensus protocols.

Safety

- The system does not produce conflicting results.
- Honest nodes will not accept two different versions of the blockchain.
- Example: A transaction cannot be both “confirmed” and “rejected” at the same time.

Liveness

- The system continues to make progress.
- Valid transactions will eventually be confirmed.
- Prevents the blockchain from getting stuck.

Finality

- Once a transaction is finalized, it cannot be reversed.
- Gives users confidence that payments are permanent.
- Important for financial systems and smart contracts.

In simple terms:

- Safety = no wrong decisions
 - Liveness = system keeps running
 - Finality = decisions cannot be changed
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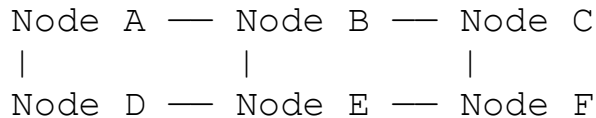
3. Consensus in P2P Networks (Visual Topology Explanation)

Blockchain operates on a peer-to-peer (P2P) network.

P2P Network Characteristics

- No central server
- All nodes are equal peers
- Nodes communicate directly with each other

Visual Topology (Conceptual)



How Consensus Works in P2P

1. A node broadcasts a transaction.
2. Other nodes verify it independently.
3. Nodes share blocks and validation results.
4. Consensus rules decide which block is accepted.

4. Common Attack Models in Consensus Systems

Consensus mechanisms must defend against attacks.

1. Double Spending Attack

- An attacker tries to spend the same coin twice.
- Consensus prevents this by agreeing on transaction order.

2. 51% Attack

- An attacker controls more than 50% of the network's power.
- Can manipulate transaction history.
- More difficult in large, decentralized networks.

3. Sybil Attack

- One attacker creates many fake nodes.
- Tries to gain influence in consensus.
- Prevented by mechanisms like Proof of Work or Proof of Stake.

4. Network Partition (Eclipse Attack)

- A node is isolated from the real network.
 - Receives false information.
 - Consensus protocols assume partial network failures.
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